ACS APPLIED MATERIALS & INTERFACES

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Scalable Synthesis of $Pt/SrTiO_3$ Hydrogenolysis Catalysts in Pursuit of Manufacturing-Relevant Waste Plastic Solutions

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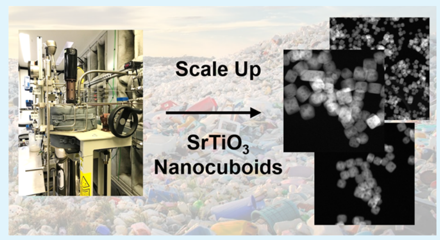
Cite This: *ACS Appl. Mater. Interfaces* 2021, 13, 58691−58700 Read Online

ABSTRACT: An improved hydrothermal synthesis of shape-controlled, size-controlled 60 nm $SrTiO_3$ nanocuboid (STO NC) supports, which facilitates the scalable creation of platinum nanoparticle catalysts supported on STO (Pt/STO) for the chemical conversion of waste polyolefins, is reported herein. This synthetic method (1) establishes that STO nucleation prior to the hydrothermal treatment favors nanocuboid formation, (2) produces STO NC supports with average sizes ranging from 25 to 80 nm with narrow size distributions, and (3) demonstrates how $SrCO_3$ formation and variation in solution pH prevent the formation of STO NCs. The STO synthesis was scaled-up and conducted in a 4 L batch reactor, resulting in STO NCs of comparable size and morphology (m = 22.5 g, d_{avg} = 58.6 ± 16.2 nm) to those synthesized under standard hydrothermal conditions in a lab-scale 125 mL autoclave reactor. Size-controlled STO NCs, ranging in roughly 10 nm increments from 25 to 80 nm, were used to support Pt deposited through strong electrostatic adsorption (SEA), a practical and scalable solution-based method. Using SEA techniques and an STO support with an average size of 39.3 ± 6.3 nm, a Pt/STO catalyst with 3.6 wt % Pt was produced and used for high-density polyethylene hydrogenolysis under previously reported conditions (170 psi H_2, 300 °C, 96 h; final product: M_w = 2400, $Đ$ = 1.03). As a well-established model system for studying the behavior of heterogeneous catalysts and their supports, the Pt/STO system detailed in this work presents a unique opportunity to simultaneously convert waste plastic into commercially viable products while gaining insight into how scalable inorganic synthesis can support transformative manufacturing.



KEYWORDS: *heterogeneous catalysis, nanomaterials, hydrogenolysis, upcycling, waste plastic, circular economy, advanced manufacturing, perovskites*

INTRODUCTION

Owing to their critical importance in numerous industrial processes, heterogeneous catalysts and the reactions they catalyze remain one of the most actively studied areas of modern scientific research.[1−3] Within this field, catalyst supports have been closely studied for their ability to improve catalytic performance. These supports, typically metal oxides such as alumina or silica, are often selected for desirable physical properties such as thermal and chemical stability; they also often stabilize active catalysts on their surfaces, thus minimizing or eliminating detrimental processes such as sintering.[1−3] Thus, designing catalysts with superior performances requires careful investigation of both catalyst behavior, support behavior, and catalyst−support interactions.[4−8] In pursuit of this, model catalyst systems, with features such as high-surface-area supports and clear support faceting, offer a unique opportunity to study as-yet unexplored systems in route to rational catalyst design.[1−3] Currently, there is significant interest in studying single-use plastic hydrogenolysis, given acute worldwide concern over plastic waste pollution. Catalytic hydrogenolysis, which can upcycle waste polyolefins into value-added products, is an attractive method by which to both mitigate the growing problem of plastic waste and utilize an untapped resource en route to a circular economy (Figure 1).[9−22]

To that effect, Celik et al. and Hackler et al. recently investigated a system of platinum (Pt) nanoparticles deposited onto $SrTiO_3$ (STO) nanocuboids (Pt/STO) that catalyze the hydrogenolysis of high-density polyethylene (HDPE) into lubricant materials.[21,22] In addition to demonstrating superior catalytic performance to a conventional catalyst, Pt on γ-Al_2O_3, several properties of Pt/STO make it an attractive model system for catalytic study. STO nanocuboids are highly crystalline, have high thermal stability, and have well-defined {100} facets, among other properties.[21,23−25] Most importantly, these {100} facets exhibit a close interfacial energy match with FCC Pt, which leads to epitaxial stabilization of the deposited Pt metal.[8,26,27] For these reasons and others, Pt/STO catalysts have been used to catalyze a variety of reactions such as propane

Received: September 27, 2021
Accepted: November 21, 2021
Published: December 2, 2021

https://doi.org/10.1021/acsami.1c18687
ACS Appl. Mater. Interfaces 2021, 13, 58691−58700

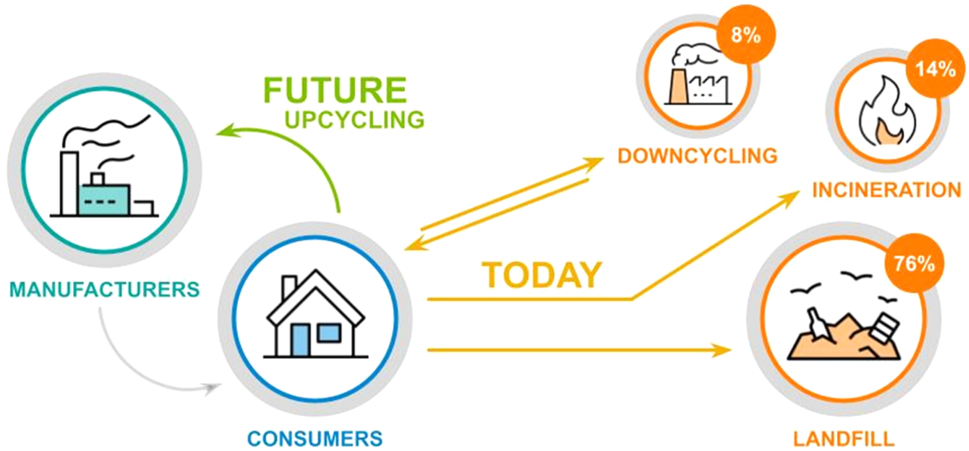


Figure 1. Current and future routes for addressing plastic waste. Currently, most plastics are lost to landfills (76%), while the rest are either incinerated (14%) or downcycled (8%) to less-valuable materials. In the future, upcycling processes could redirect up to 100% of postconsumer materials back to manufacturers.[9]

oxidation and oxidative coupling of methane.[26−30] Because catalytic hydrogenolysis of polymers is both largely unexplored and potentially possesses wide-ranging economic and ecological benefits, there is a strong incentive to continue studying both the Pt/STO catalyst and the STO support itself.

Previously, Rabuffetti et al. established the superiority of hydrothermal synthesis over molten-salt and solid-state methods in producing STO NCs.[31] Based on that work, our group has explored faceting, surface reconstructions, and catalyst/support interactions of a range of titanate perovskites through the effect of metal precursors, surface directing agents (changing both surface termination and faceting), and microwave hydrothermal synthesis.[32−37] Other approaches have been used to control the morphology of titanate perovskite nanoparticles.[38−43] However, these nanoparticle syntheses can be difficult to reproduce, as the nucleation and growth processes can be highly condition-, lab-, and researcher-specific, in ways that are difficult to identify and quantify.[38−41]

Moreover, increasing the synthetic output of STO NCs in pursuit of a commercially viable upcycling catalyst will generate additional, scale-specific challenges. First, there are several considerations that inherently complicate advanced manufacturing processes, including but not limited to (1) demonstrating atomic-level spatial and compositional control of materials while producing at manufacturing-relevant scales; (2) tailoring processes to meet machining constraints, and (3) utilizing pathway engineering to prevent the formation of undesirable design modalities, e.g., nontarget particle morphologies. Furthermore, several challenges specific to lubricant production must also be considered. For example, firms often have product-specific demands for plastic feedstocks that are not shared broadly, creating a demand for varied raw materials with differing yet equally narrow specifications. Furthermore, switching to recycled feedstocks will likely also require retooling factory equipment to account for feedstock property changes and thereby reduce the economic incentive for the use of recycled materials. In total, these myriad concerns regarding both catalyst and lubricant production demonstrate the need for a catalyst synthesis method that exhibits control over STO NC physical properties (i.e., surface area, particle morphology), retains the potential to be scaled in a straightforward manner, and produces a highly uniform upcycling product that can meet stringent manufacturer's specifications.[44,45]

Herein, we report an STO hydrothermal synthesis that identifies parameters whose control is critical to the formation of nanocuboids, produces STO samples with at least 80% NCs, and explains the impact of secondary phases such as $SrCO_3$ on the formation of the primary STO NC phase. The order of addition of solutions during the creation of the Sr−Ti−OH prehydrothermal mixture has been investigated to determine how it can lead to the formation of either NCs or irregularly shaped nanoparticles. Through the analysis of prehydrothermal Sr−Ti−OH particles, the dependence of the hydrothermal formation of STO NCs on both the phase and size of the precursor particles was also noted. Additionally, shortening the Sr−Ti−OH mixture stir time was shown to potentially decrease $SrCO_3$ formation. These modifications were incorporated into the synthetic method, and STO NCs with average sizes ranging from 25 to 80 nm with equivalent nanocuboid morphologies were synthesized by modifying the concentrations of reagents. In pursuit of synthetic scale-up to industrial scale, these STO supports were synthesized in a 4 L batch reactor. The resultant STO NCs are comparable in average size and morphology to those synthesized herein in a 125 mL autoclave reactor, suggesting that with the modifications reported in this work, a further scale-up of the reaction is likely attainable.

Due to the inherent limitations of scaling up atomic layer deposition, which Celik et al. and Hackler et al. utilized to generate Pt/STO catalysts,[21,22] solution-phase deposition methods were explored to find a process that can both retain some of the precision of atomic layer deposition and provide a clear path toward use on a manufacturing-relevant scale. To that effect, Pt/STO catalysts were synthesized through strong electrostatic adsorption (SEA), a charge-mediated, solution deposition method capable of depositing small Pt nanoparticles in a controlled fashion.[46,47] These Pt/STO catalysts have been tested for HDPE hydrogenolysis and produced wax products that, while larger than those observed by Celik et al., have very low dispersity and high uniformity.

■ EXPERIMENTAL SECTION

Materials and Instrumentation. *Powder X-ray Diffraction.* Diffraction experiments were conducted on a Rigaku Ultima powder X-ray diffractometer (PXRD) and analyzed using JADE software (Materials Data, Inc.). In preparation for analysis, solid samples were ground with a mortar and pestle. Gel aliquots were collected and spread

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ACS Appl. Mater. Interfaces 2021, 13, 58691−58700

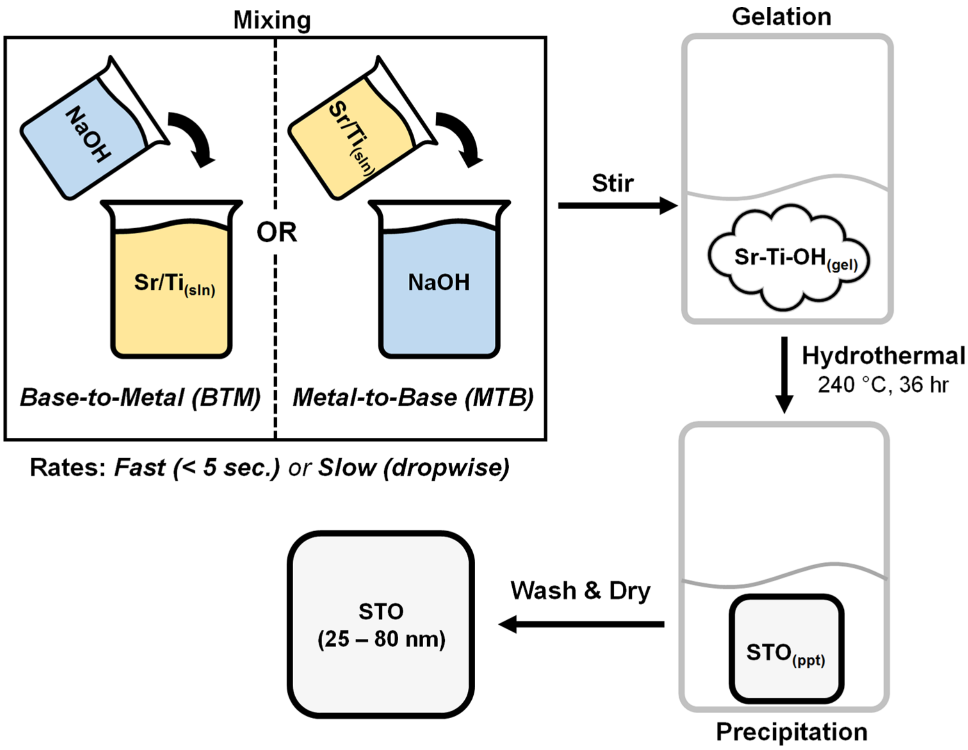

Figure 2. Schematic of STO NC synthesis and possible orders of operation for Sr−Ti−OH gel creation. The outcome of the STO synthesis is highly sensitive to the base and metal solution mixing step. Sr−Ti−OH gels can be created either via metal-to-base (MTB) or base-to-metal (BTM) addition.

flat on the PXRD sample holder. The crystallite size was obtained via peak broadening analysis in MDI JADE. Peak broadening was calculated via the Scherrer equation for each peak in the diffraction pattern, and the average of sample values was taken.

Electron Microscopy. Nanoparticle imaging was conducted by transmission electron microscopy using a Hitachi H8100 TEM, a Hitachi HD2300, and an FEI Talos F200X TEM/STEM, all operated at 200 kV. These microscopes were used in coordination with the NU Atomic and Nanoscale Characterization Experimental Center at Northwestern University and the Center for Nanoscale Materials at Argonne National Laboratory. In preparation for analysis by electron microscopy, gel and solid samples (~20 mg) were sonicated in ethanol (10 mL) for 15 min. The resulting suspension was dropcast onto a lacey carbon TEM grid (Ted Pella, Inc., UC-A on Lacey 400 mesh Cu).

Electron Micrograph Analysis. Particle size and shape were measured using ImageJ and Gatan Digital Micrograph software.[48] The face-to-face distance between the opposing [100] faces of STO nanocuboids was used as a measure of particle width. Particle rounding was measured by inscribing a circle in the corner of an STO nanoparticle such that the circumference of the resulting circle traced the nanoparticle corner.[49] The radius of this circle was then used as a means of comparing corner rounding (Supporting Information Figure S1).

HDPE Hydrogenolysis. A Parr reactor and a high-throughput screening pressure reactor (SPR; Unchained Labs) at the Argonne National Laboratory's High-Throughput Research Laboratory were used for catalytic activity experiments. Activity experiments were performed under solvent-free conditions at 170 psi and 300 °C, unless otherwise noted. For the Parr reactor, the conditions were 300 mg Pt/STO, 3 g HDPE, Sigma-Aldrich, M_w = 35 kDa, $Đ$ = 3.11, 300 °C, 170 psi H_2, 96 h, with a glass liner in the reactor, as per Hackler et al.[22]

Gel Permeation Chromatography (GPC). Samples were analyzed by GPC as per Hacker et al.[22] HQL samples from hydrogenolysis of HDPE were analyzed for molecular weight (M_n and M_w) and molecular weight distribution (M_w/M_n) by high-temperature gel permeation chromatography (Agilent-Polymer Laboratories 220). 1,2,4-Trichlorobenzene (TCB) containing 0.01 wt % 3,5-di-*tert*-butyl-4-hydroxytoluene (BHT) was chosen as the eluent at a flow rate of 1.0 mL/min at 150 °C. Lubricant samples were prepared in TCB at a concentration of approximately 2.0 mg/mL and heated at 150 °C for 24 h prior to injection.

Note: All glassware was washed thoroughly with a weakly acidic (<20 wt %) HCl solution and deionized water prior to use. Stir bars were stored in a weakly acidic HCl solution and thoroughly rinsed with deionized water prior to use.

Synthesis. *Precursor Solutions. Solution A.* As-received $Sr(OH)_2 \cdot 8H_2O$ (2.55 g, 10.0 mmol, Sigma-Aldrich, 99.5%) was added to acetic acid (2.86 mL, 0.05 mol, Sigma-Aldrich, 99.5%) in deionized water (20−60 mL) and stirred for 10 min.

Solution B. $TiCl_4$ (1 mL, 9.1 mmol, Sigma-Aldrich, 99.5%) was added to ethanol (20 mL, absolute) in a 50 mL beaker via a Luer Lock syringe and stirred for 10 min.

Hydrothermal Synthesis. Solutions **A** and **B** were mixed in a 100 mL Teflon beaker and stirred for approximately 10 min to ensure homogeneity (**solution AB**). **AB** was combined with NaOH (10 M or saturated) by the addition of base to metal (BTM, i.e., introduction of NaOH into **AB** to form **AB−OH**) or the addition of metal to base (MTB, introduction of **AB** into NaOH to form **OH−AB**) depending on the experiment. A mixture temperature of 35−55 °C was observed immediately after a combination of the bimetallic solution and NaOH. The sample was stirred for 10 min on a stir plate and transferred to a 125 mL autoclave. The autoclave was treated in a Carbolite laboratory oven (1 °C/min ramp rate, 36 h, 240 °C) and then allowed to cool ambiently to room temperature. The resulting white powder was washed repeatedly with deionized water via centrifuge (4500 rpm, 7 min) until the supernatant was pH 7 and then dried in air in an oven (80 °C, overnight).

Safety note: A combination of **AB** (pH ~ 4) and **OH** (pH ~ 14) generates heat and increases the resultant reaction mixture by 10−30 °C. When conducting this synthesis beyond the 125 mL lab scale, slow MTB addition should be employed to prevent the rapid evolution of heat.

Large-Scale Hydrothermal Synthesis. Scale-up experiments were conducted at the Materials Engineering Research Facility, part of the Applied Materials Division of Argonne National Laboratory.

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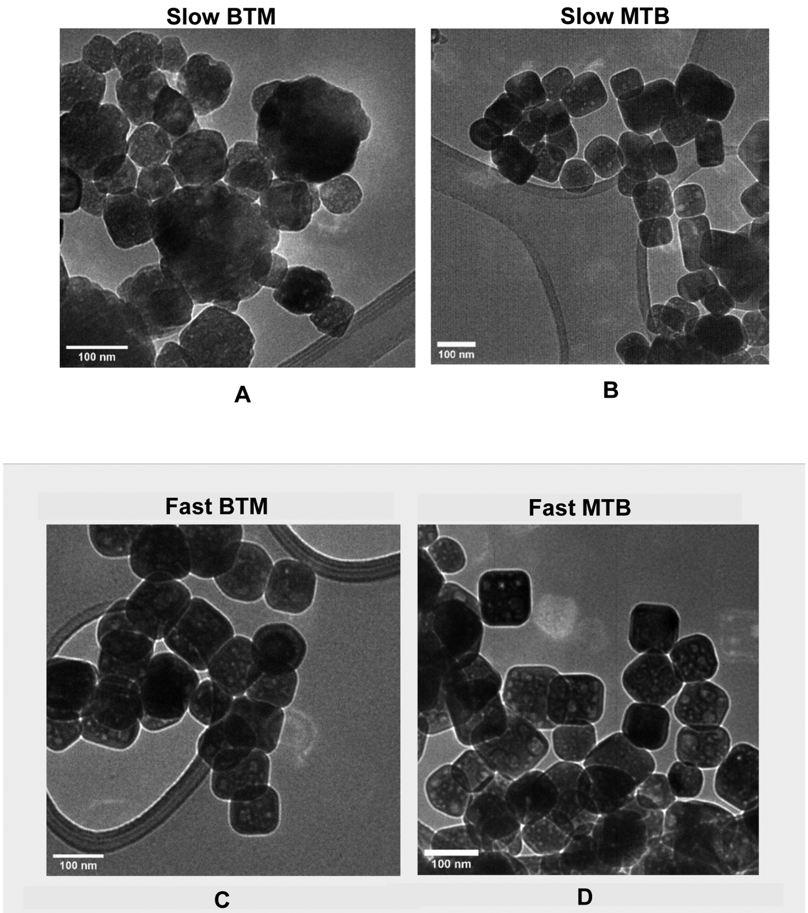

Figure 3. Transmission electron microscopy-bright field (TEM-BF) micrographs of STO nanocuboids synthesized from each of four possible gel creation conditions. Each of four conditions was established through variation of the addition rate (fast or slow, i.e., combination of components in either <5 s or dropwise) and the order in which components are combined (base to metal or metal to base).

The solution concentrations were proportionally scaled for a 4 L reactor, with the concentrations of Sr, Ti, and NaOH halved relative to the liquid volume to decrease the concentration of chloride ions in the unlined Hastelloy C-276 reactor.

Solution **A**: 38.69 g of $Sr(OH)_2 \cdot 8H_2O$ was dissolved in a solution of 48.04 g of acetic acid and 640.00 g of H_2O under stirring for 2 h.

Solution **B**: 27.62 g of $TiCl_4$ was dissolved in 504.96 g of ethanol for 10 min.

Solutions **A** + **B** were mixed and stirred for 10 min in a 2000 mL Erlenmeyer flask; 276.85 g of 10 M NaOH solution was injected into the **AB** solution for 28 min at a flow rate of 10 mL/min using a syringe pump. The **AB**−**OH** solution was stirred using a magnetic stirrer at the maximum stir rate of the stir plate; no gelation was observed. After mixing and stirring, the **AB**−**OH** solution was allowed to sit for 10 min; the pH of the solution was 13.0, and sedimentation occurred.

The solution and sediment were transferred to a 4 L Hastelloy C-276 reactor with an internal impeller and a heating jacket. The reactor was heated to 240 °C at a rate of 2 °C/min and held for 12 h with a stir speed of 400 rpm. After the reaction, the reactor was cooled at a rate of 2 °C/min.

The precipitate and liquid were transferred from the reactor to a secondary container. The solution pH was 13.2 after the hydrothermal reaction. Initially, the liquid was a light yellow color but changed to dark orange with time. The precipitate was washed and dried.

Gel Precursor. **AB**−**OH** or **OH**−**AB** was prepared and stirred for 50 min, while the pH of the solution was measured with pH paper at 5 min intervals. The resulting product was centrifuged once (4500 rpm, 5 min), and the supernatant was decanted. A white gel was recovered and immediately prepared for analysis by either PXRD or TEM to prevent decomposition.

Pt Deposition. Pt nanoparticles were deposited via strong electrostatic adsorption, a charge-mediated solution process.[46,47] $Pt(NH_3)_4(NO_3)_2$ (PTA, Sigma-Aldrich, 60 mg) was added to a 15 mL glass vial containing a stir bar and STO nanocuboids (0.4 g) synthesized herein. Next, a NaOH solution (3 mL, pH 11) was added to the vial and the resulting mixture was stirred for 2 h. The suspension was then allowed to settle, the supernatant was decanted, and the resulting product was dried in air overnight. This ligated Pt/STO was reduced in a tube furnace (5% H_2/N_2, 450 °C, 12 h) to afford the final Pt/STO catalyst.

■ RESULTS AND DISCUSSION

Batch-to-batch variation in size, morphology, and phase purity when following the synthesis described by Rabuffetti et al. led us to focus on steps in the synthesis that could lead to such a variation. The metal and base mixing step (Figure 2) were identified as the step most likely to be sensitive to small changes. Thus, a series of modifications were made to improve the reliability of the synthesis by increasing control over the said step. These changes result in a synthesis that produces >80% nanocuboids. First, 5 g of NaOH pellets was replaced by a 13 mL, 10 M NaOH solution to facilitate the combination of NaOH (**OH**) and the bimetallic Sr/Ti mixture (**AB**) (**AB**−**OH**). Distinct orders of operations were then established for the combination of **AB** and **OH** to identify all possible pathways by which the **AB**−**OH** gel could be created (Figure 2).

These pathways were defined as follows: addition rates were labeled either fast (i.e., component is added in under 5 s) or slow

https://doi.org/10.1021/acsami.1c18687
ACS Appl. Mater. Interfaces 2021, 13, 58691−58700

(component is added dropwise over the course of stirring), while the orders of addition were labeled either base to metal (BTM, NaOH is added into Sr/Ti) or metal to base (MTB, Sr/Ti is added into NaOH). **AB−OH** was synthesized under each of four possible pathways and treated hydrothermally (240 °C, 36 h).

Electron micrographs of the resultant STO nanoparticles are presented in Figure 3. From these micrographs, it is apparent that precursor gels synthesized under slow BTM conditions (Figure 3a) afford irregularly shaped nanoparticles, while gels synthesized under all other conditions (Figure 3b−d) afford rounded nanocuboids, with yields up to 1.5 g.

To understand the driving forces behind variations in STO morphology, each **AB−OH** gel precursor was isolated and analyzed at 5 min intervals of mixing without hydrothermal treatment. The corresponding solution pH values are presented in Figure 4 and display two distinct environments: one in which

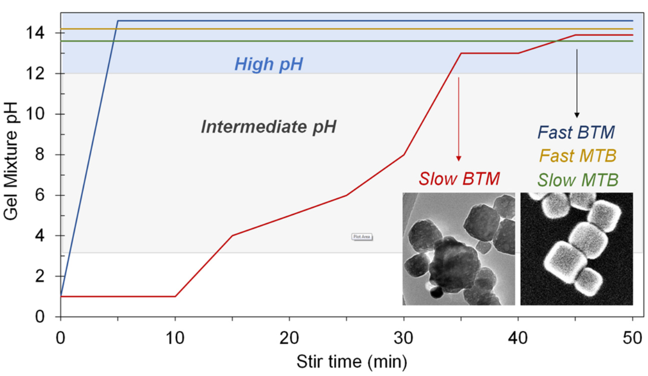

Figure 4. Rate of pH change in STO precursor gels based on the order of operation. Gels synthesized via fast BTM, slow MTB, and fast MTB pathways show a rapid jump to pH ∼ 14, while gels synthesized via the slow BTM pathway increase over the duration of mixing from pH ∼ 1 to ∼14. Depending on specific reaction conditions, all temperature samples experience a temperature increase of 10−30 °C between the start of mixing and complete incorporation of NaOH. In alkaline titanate systems, the perovskite STO phase is the most stable phase in the "high pH" region, shown here in light blue (pH 12−14).[50−54]

the pH of **AB−OH** is 14 from the start of stirring ("constant pH") and one in which the pH increases in a stepwise fashion, reaching 14 after all NaOH has been added ("climbing pH"). This suggests that STO NC synthesis requires rapid formation of a pH 14 environment (vide infra).

Next, **AB−OH** mixtures made via all four pathways were analyzed by powder X-ray diffraction. Table 1 reports the phases observed in each gel environment, final particle morphology after hydrothermal treatment, and average crystallite size within the gel. In the case of fast BTM, fast MTB, and slow MTB additions, an STO primary phase (94−99%) and less significant $SrCO_3$ secondary phase (1−6%) are observed. The average crystallite size in these environments is also comparable, ranging from 21.0 nm to around 31.3 nm. By contrast, Sr−Ti−OH gels created in a slow BTM condition only show crystalline phases above pH 13, at which point a $SrCO_3$ primary phase (64%) and an STO secondary phase (36%) are observed. The average crystallite size in this sample is significantly smaller at an average of 10.7 nm. Diffraction patterns corresponding to each entry in the table are presented in the Supporting Information (Figure S2).

Table 1. Phase Identification in Prehydrothermal STO Colloidal Precursors

gel environment	% STO	% $SrCO_3$	average STO crystallite size (nm)	posthydrothermal morphology
slow BTM	36	64	10.7 ± 5.5	irregularly shaped
fast BTM	94	6	25.2 ± 7.3	cuboid
slow MTB	97	3	21.0 ± 1.9	cuboid
fast MTB	99	1	31.3 ± 9.1	cuboid

The order of the addition of the metal and base solutions during the gel creation step has a significant effect on the precursor particle formation (Table 1) and the subsequent hydrothermal formation of STO NCs (Figure 3). The optimum order, which maximizes STO NC formation and minimizes $SrCO_3$ in the precursor, is the fast MTB method. A significant percentage of the STO nanoparticles were NCs for the fast BTM and slow MTB methods, although higher percentages of $SrCO_3$ were observed in those precursor solutions than in the fast MTB method. This is consistent with the original method reported by Rabuffetti et al., which specified pouring NaOH pellets into the Sr/Ti solution, approximately equivalent to the fast BTM addition.

The irregular STO nanoparticles resulting from the slow BTM method are significantly different from the STO nanoparticles from the other methods (Figure 3). Where this method differs is in how rapidly the solution pH increases, with respect to the local environment of the Sr and Ti ions. For the slow BTM method, the pH gradually increases from ∼1 to ∼14 with the addition of NaOH, while for all other methods, the pH rapidly jumps from 1 to 14 (Figure 4). This is significant, as the stability of various ions, complexes, and solid phases change with pH.[52−54] At low pH, Sr and Ti ions and complexes are stable in solution. At intermediate pH, Ti will condense as amorphous sol−gels, which are not observable by PXRD.[55] At high pH (12−14), STO is stable and will crystallize out of solution.[54] If there is a significant amount of CO_2 dissolved in solution, this may compete with STO to form $SrCO_3$ at high pH.[52,53] Significantly, if the concentration in the solution of Sr and Ti is not stoichiometric, Sr or Ti may form secondary phases instead of STO, such as $SrCO_3$, $Sr(OH)_2$, or TiO_2, depending on whether there is a higher concentration of Sr or Ti. In the case of the slow BTM method, the long dwell time at intermediate pH allows the formation of titania sol−gels, which pulls Ti out of solution. When the solution pH finally reaches the point that STO is stable, the solution is Ti-deficient and the Sr-rich $SrCO_3$ phase forms in addition to STO (Table 1).

Particles from **AB−OH** solutions from both constant pH and climbing pH environments were imaged by electron microscopy to observe possible morphological differences in the Sr−Ti−OH precursor under varied conditions; corresponding electron micrographs are presented in Figure 5. Precursor gels prepared in a constant pH environment form aggregate nanoparticles with rough surfaces similar in size to the hydrothermally treated STO, while gels prepared in a climbing pH environment form an enmeshed, irregular, fiberlike morphology that does not resemble STO nanoparticles. Elemental mapping of the gel precursors in Figure 5 shows generally homogeneous distributions of strontium and titanium in the STO gel precursor irrespective of the pH environment in which it was prepared (Figure S3).

Next, the concentrations of [Sr], [Ti], and [OH] in the Sr−Ti−OH mixture were varied to determine their influence on the final STO particle size. Specifically, the molarity of the 13 mL

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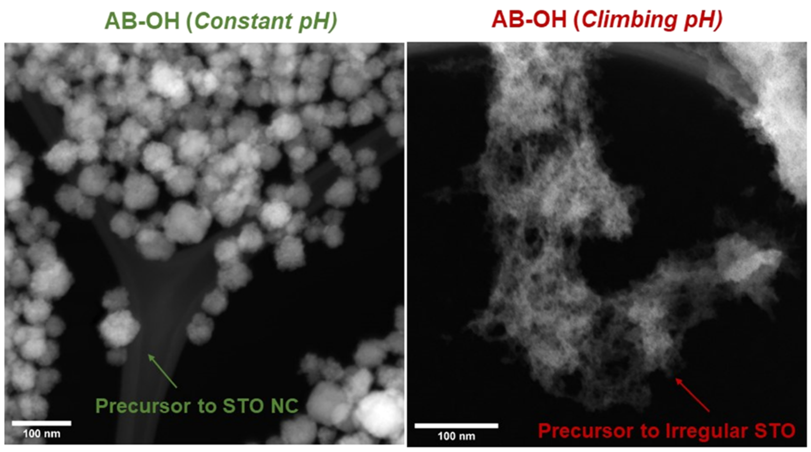

Figure 5. TEM-high-angle annular dark-field (HAADF) micrographs of STO gel precursor in constant pH (left, green) and climbing pH conditions (right, red). Images presented are representative of each gel precursor solution throughout the duration of mixing.

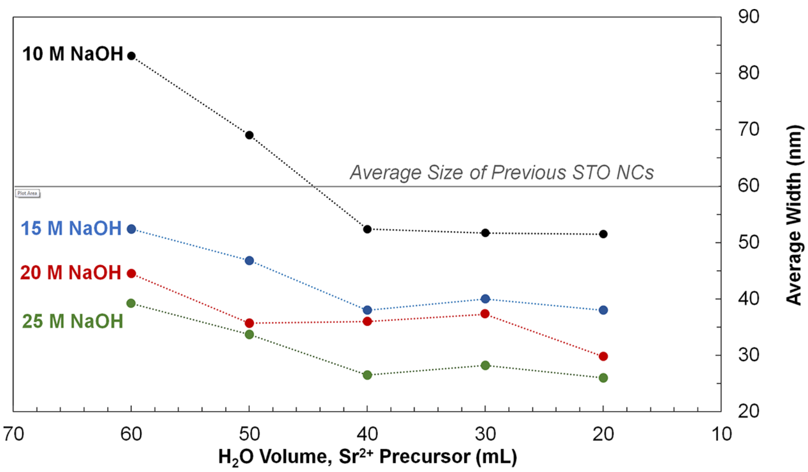

Figure 6. Average sizes of STO nanocuboids produced through variation of added H_2O and NaOH. Variation of the molarity of added NaOH and water volume used in the reaction allows for the synthesis of STO NCs ranging from 25 to 80 nm.

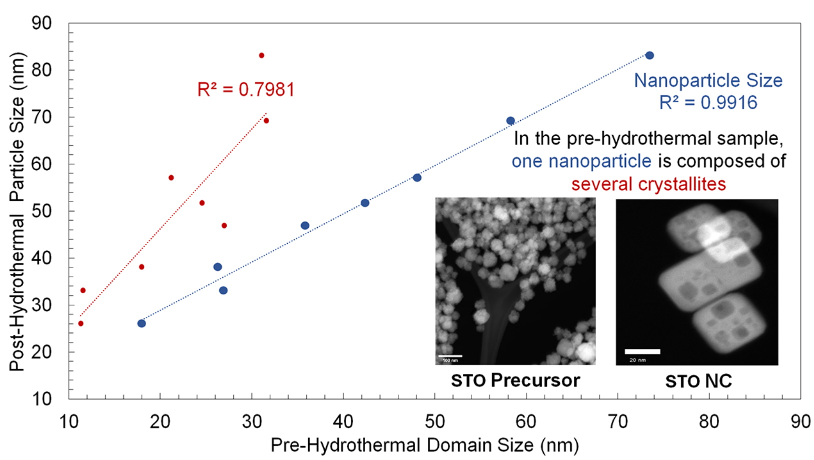

Figure 7. Size measurements for gel precursors compared to the final STO nanocuboid size. The average size of STO crystallites (red, PXRD) and particles (blue, EM) from the precursor gel is compared against the average STO NC size (TEM-HAADF) after hydrothermal treatment. There is a correlation between both precursor crystallites and particle sizes and the final nanocuboid size.

NaOH was varied, as was the amount of water used to dissolve 2.55 g $Sr(OH)_2 \cdot 8H_2O$ in **solution A**. The results of these experiments are presented in Figure 6. STO nanocuboids ranging from 25 to 80 nm are attainable when varying the volume of water from 20 to 60 mL and varying the amount of added NaOH from 13 mL at 10 M to 13 mL at 25 M. Both the concentration of the added base and the volume of water added have significant effects on the final particle size. Global synthetic conditions (i.e., values of [NaOH] and water volume measured at the point of incorporation into the reaction mixture) and

https://doi.org/10.1021/acsami.1c18687
ACS Appl. Mater. Interfaces 2021, 13, 58691−58700

corresponding average particle sizes for each data point are reported in Figure S4a, along with a graphical representation of the average size and variance in Figure S4b. Several representative micrographs are also reported in the Supporting Information (Figure S4c).

To probe whether particle morphology was consistent across each sample, an average measure of corner rounding was obtained by inscribing a circle into the corner of each nanocuboid, the radius of which was used as a measure of curvature (Figure S1a). The data show that the ratios of particle size to corner rounding (d:r) are comparable, ranging from 4.0 ± 1.3 to 4.9 ± 1.3. Based on these values, the STO particle morphology appears generally equivalent across all samples (Figure S1b,c).[49]

The average aggregate particle size and crystallite size were measured for precursors synthesized under constant pH conditions, corresponding to STO NCs ranging from 25 to 80 nm (Figures 7 and S5). There is a correlation between both the crystallite and particle aggregate size of the precursor particles and the size of the final STO NCs, with larger crystallites and aggregates forming larger NCs.

When designing a catalyst support, it is desirable to maximize the surface area while retaining beneficial structural properties (e.g., faceting), increasing the potential number of catalytic sites. Particle size, which correlates inversely with surface area, can be varied over the range of 25−80 nm without significantly affecting STO NC morphology through control of the base and total ion concentrations (Figures 6 and S1). Both the concentration of base ([NaOH]) and the total ion concentration (controlled by the volume of water) affect the average particle size in the precursor and the STO NC (Figure 7), with higher ionic concentrations (base or total) decreasing the particle size. In the range of conditions probed, the concentration of base has a more significant effect on STO NC size, but the trend in the 10 M NaOH samples at the highest water volumes (lowest total ion concentrations) suggests that decreasing the total ion concentration further may significantly increase the STO NC size. For the precursors that form STO NCs, there is a strong correlation between the precursor crystallite (PXRD) and particle (EM) size and the final STO NC size (Figure 7), which suggests that the NCs form directly from the precursor particles with minimal diffusion between particles. That the precursors primarily composed of $SrCO_3$ and amorphous titania (from the slow BTM method) do not form STO NCs under the tested conditions is likely because the rates of precursor dissolution, diffusion, and STO formation are overall slower than the rate of STO recrystallization for the aggregates of STO crystallites. Based on these data, STO nanocuboid formation likely proceeds through separate synthesis-dependent nucleation and growth process. Initial STO nucleation at high pH likely occurs in a rapid burst fashion governed by thermodynamic and/or colloidal stability, followed by diffusion-limited facet growth during hydrothermal treatment. We believe that changing precursor concentrations (and therefore Sr^{2+}, Ti^{4+}, and OH^- concentrations) affects the minimum stable crystallite size that can form upon the creation of the Sr−Ti−OH gel mixture. More specifically, lower precursor concentrations likely decrease mixture saturation and increase the minimum stable crystallite size required for STO nanoparticles to form without redissolving.

As observed in the slow BTM synthesis, the formation of large amounts of $SrCO_3$ negatively affects the formation of STO NCs. $SrCO_3$ may also form by the adventitious dissolution of CO_2 in solution[56] or through variation in the Sr/Ti ratio. These effects and methods to mitigate them have also been explored. In samples in which $SrCO_3$ formation is observed through powder X-ray diffraction, the final STO nanoparticles have irregular morphologies (Figure S6). Sr−Ti−OH mixture time was reduced from 50 to 10 min, as this appeared to reduce the likelihood of $SrCO_3$ formation. Complete elimination of $SrCO_3$ formed from adventitious CO_2 dissolution would likely require the use of an air-free environment or degassing of the water−ethanol mixture.[41]

Furthermore, we have observed that the global Sr/Ti ratio (defined based on the amounts of $Sr(OH)_2{\cdot}8H_2O$ and $TiCl_4$ added to the precursor solutions) significantly impacts nanoparticle morphology. Samples with a Sr/Ti ratio of 1.05:1 most consistently form STO NCs, while samples with higher (>1.1:1) Sr/Ti ratios form noncubic nanoparticles (Figure S7a). Previously, samples with both NC and irregular morphologies had been observed from STO syntheses using a global Sr/Ti ratio of 1:1. This could be due to the decomposition of the $Sr(OH)_2{\cdot}8H_2O$ precursor from the octahydrate phase to less hydrated phases, which was likely accelerated through grinding of the as-received reagent and storage in a desiccator (Figure S8a). As such, this updated synthetic method calls for using as-received $Sr(OH)_2{\cdot}8H_2O$.

Preliminary scale-up experiments have been conducted and have demonstrated that STO nanocuboids can be synthesized at scales larger than a 125 mL autoclave. A highly cubic STO sample (m = 22.5 g) was synthesized in a 4 L batch reactor at the Materials Engineering Research Facility (MERF) at Argonne National Laboratory; this is a 32-fold scale-up from the hydrothermal conditions reported by Rabuffetti et al. TEM micrographs of the resultant STO NCs are displayed in Figure 8.

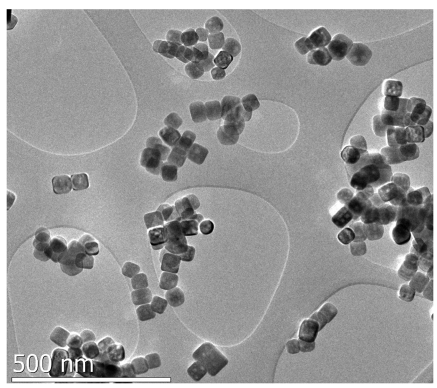

Figure 8. TEM-BF micrographs of STO nanocuboids synthesized in a 4 L batch reactor. This sample (m = 22.5 g) has an average size of 58.3 ± 16.2 nm. The STO NC morphology of these nanoparticles is generally equivalent to those samples synthesized in a 125 mL autoclave.

These NCs have an average size and variance of 58.3 ± 16.2 nm, with an average corner rounding of 10.5 ± 1.7 nm, resulting in a d:r of 5.5: 1. This is beyond the range of d:r ratios seen in the highly cubic samples synthesized in a 125 mL autoclave, but within error of the higher d:r values from that group of samples, and therefore, not indicative of a change in the Wulff shape under hydrothermal conditions. These corner rounding measurements are all also comparable to an equivalent measurement obtained for a sample synthesized by Rabuffetti et al. with the previous hydrothermal method (Figure S1b; d:r = 5.0 ± 1.2).

https://doi.org/10.1021/acsami.1c18687
ACS Appl. Mater. Interfaces 2021, 13, 58691−58700

These preliminary results show that scale-up from a 125 mL autoclave reactor to a 4 L autoclave reactor can be attained without significantly altering the average nanoparticle size and shape, a fact supported by the average size d and average ratio of size to corner rounding, d:r. Thus, it is likely possible to further scale-up the STO hydrothermal synthesis described herein while retaining highly cubic particle morphologies.

Size-controlled STO supports reported above have been used to construct Pt/STO catalysts through strong electrostatic adsorption (SEA), a charge-mediated, aqueous method adapted for depositing Pt onto the STO surface. Specifically, an STO nanocuboid sample with an average size of 39.3 ± 6.3 nm was used to deposit small Pt nanoparticles (<2 nm; Figure 9) with an

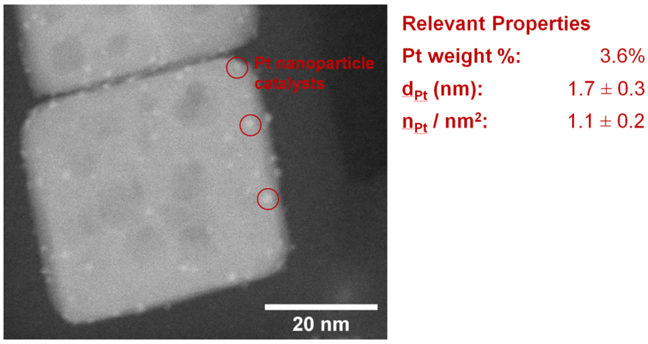

Figure 9. TEM-HAADF micrograph of SEA-derived Pt/STO catalysts. This sample was synthesized via the updated hydrothermal synthesis and SEA deposition process reported herein. This sample has a Pt weight loading of 3.6%, an average Pt particle size of 1.7 ± 0.3 nm, and an average Pt area loading of 1.1 ± 0.2 Pt atoms per square nanometer.

average diameter of 1.7 ± 0.3 nm, an average area loading of 1.1 ± 0.3 Pt atoms/nm^2, and a weight loading of 3.6 wt % Pt. The successful deposition of Pt onto the STO NC surface suggests that the improvements made to the STO hydrothermal synthesis reported herein do not significantly alter the support surfaces, thereby making them generally equivalent to samples synthesized via the hydrothermal synthesis reported by Rabuffetti et al. Furthermore, these results indicate that the SEA process is (1) a precise deposition method that deposits Pt onto STO NCs in a controlled fashion and (2) amenable to scale-up due to the comparatively simple equipment required to conduct the deposition, as compared to other surface-limited processes like ALD.

The above SEA-Pt/STO sample was tested for the hydrogenolysis of HDPE using the conditions described by Hackler et al.[22] After 96 h with 11 mg of 1.7 nm Pt particles, the HDPE was converted from M_w = 35 kDa, $Đ$ = 3.11 to an M_w = 2400 Da, $Đ$ = 1.03 product (99% conversion to wax, 1% gas, 0% solid and liquid), compared to similar experiments in a Parr reactor in Celik et al. with 8 mg of 2.0 nm Pt particles, which generated an M_w = 2100 Da, $Đ$ = 1.7 product after 68 h without the use of a liner. The dispersity of the product of the SEA-Pt/STO catalyst decreased below 1.1, similar to our prior work producing upcycled liquid products from polyolefins with Pt/STO.[21,22] Notably, the dispersity decreased below 1.1 while the M_w was above 2000 Da, which we have not previously observed. This narrow dispersity demonstrates that SEA-derived Pt/STO is a selective catalyst for waste plastic hydrogenolysis. The conversion to a waxlike product was higher than previously reported; no conversion to liquids or light gases occurred, despite being observed for hydrogenolysis with previous Pt/STO catalysts. We postulate that these differences may be due to differences in the amount of Pt atoms per square nanometer between samples. Celik et al. and Hackler et al. have previously observed negligible changes in the Pt particle size and distribution after hydrogenolysis for ALD Pt/STO catalysts by electron microscopy and suggested that it is due to the epitaxial stabilization of Pt on the $SrTiO_3$\{100\} facets. Based on those results, we expect that SEA-Pt/STO Pt nanoparticles will be similarly stabilized. The observed low dispersity at high molecular weights will be further investigated to determine how the changes in the Pt nanoparticle dispersion from ALD to SEA affect polymer/catalyst interactions and therefore the upcycled liquid products.

■ CONCLUSIONS

Highly cubic, size-controlled STO catalyst supports have been used to generate size-controlled Pt/STO hydrogenolysis catalysts and upcycle HDPE into high-quality liquids. Several modifications to an earlier STO hydrothermal synthesis have (1) allowed for the synthesis of STO nanocubes ranging from 25 to 80 nm in size with narrower size distributions than before; (2) ensured that nanoparticles with highly cubic morphologies are formed by eliminating competing effects of reaction byproducts such as $SrCO_3$, and (3) revealed that the rapid formation of precursor crystallites in a pH 14 solution is required for nanocuboid formation. Analysis of the prehydrothermal synthetic steps established that STO NCs rapidly nucleate prior to hydrothermal synthesis, suggesting that hydrothermal treatment is primarily responsible for NC facet growth. In the future, this STO NC hydrothermal synthesis will be studied in a microwave reactor to understand the relationship between the heating method and final STO morphology and the relationship if any between these parameters and the pH control of the Sr−Ti−OH gel precursor demonstrated in this work.

We further report that the STO hydrothermal synthesis was conducted on a 4 L batch scale to afford 22.5 g of highly cubic nanoparticles. These preliminary results demonstrate that STO can be synthesized on larger scales while retaining the average particle size and a highly cubic morphology. These findings will be used to guide the conversion of the synthesis into a continuous-flow reactor in pursuit of a high-throughput synthesis of shape-controlled, size-controlled Pt/STO catalysts. The construction of Pt/STO catalysts and their upcycling of HDPE into uniform wax products also demonstrate that the physical properties of these catalysts can be varied without adversely affecting hydrogenolysis. We will continue to explore the extent to which the variation of properties such as the support surface area impacts HDPE hydrogenolysis as we pursue a commercially viable synthesis of Pt/STO catalysts for waste plastic upcycling.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsami.1c18687.

PXRD patterns of gel precursor samples, measurements of average nanoparticle sizes, EDX mapping of gel precursors, TEM/STEM images, and reaction conditions for experiments studying $SrCO_3$ formation (PDF)

https://doi.org/10.1021/acsami.1c18687
ACS Appl. Mater. Interfaces 2021, 13, 58691−58700

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was supported as part of the Institute for Cooperative Upcycling of Plastics (iCOUP), an Energy Frontier Research Center funded by the U.S. Department of Energy (DOE), Office of Science, Basic Energy Sciences (BES). Argonne National Laboratory is operated by UChicago Argonne LLC under Contract DE-AC-02-06CH11357 for the U.S. Department of Energy, and Ames Laboratory under Contract DE-AC-02-07CH11358 is operated by Iowa State University for the U.S. Department of Energy. Initial scale-up experiments at Argonne's Materials Engineering Research Facility (MERF) were supported by the U.S. Department of Energy, Laboratory Directed Research and Development funding at Argonne National Laboratory under Contract No. DE-AC02-06CH11357. This research used resources of the Advanced Photon Source (APS) and the Center for Nanoscale Materials (CNM), U.S. Department of Energy, Office of Science User Facilities operated by Argonne National Laboratory under Contract No. DE-AC02-06CH11357. The authors acknowledge Dr. Jianguo Wen from the Center for Nanoscale Materials at Argonne National Laboratory for assistance with electron microscopy.

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